

Question 1 (12 marks) Begin a new booklet

Marks

- (a) Factorise fully $16x^3 - 2$ **2**
- (b) Consider the points $A(-3, 2)$ and $B(6, -4)$. Find the coordinates of the point $P(x, y)$ that divides the interval AB in the ratio of 2:1. **2**
- (c) Solve the inequality $\frac{2}{x+1} < 1$. **2**
- (d) Consider $f(x) = 3x^3 + 6$. Explain, using calculus, why $f(x)$ is always increasing. **2**
- (e) Differentiate:
- (i) $x \cos^2 3x$ **2**
- (ii) $x^2 \tan^{-1} x$ **2**

Question 2 (12 marks) Begin a new booklet**Marks**

- (a) Find the values of k such that $(x - 2)$ is a factor of the polynomial
 $P(x) = x^3 - 2x^2 + kx + k^2$. **2**
- (b) Find correct to the nearest degree the acute angle between the lines
 $y = 3x + 1$ and $x + y - 5 = 0$ **2**
- (c) Evaluate $\int_0^1 \frac{dx}{\sqrt{4-x^2}}$ **2**
- (d) Use the substitution $t = \tan \frac{x}{2}$ to show that $\frac{1 - \cos x}{\sin x} = \tan \frac{x}{2}$ **3**
- (e) Graph $y = 2 \sin 3x$ for $0 \leq x \leq 2\pi$ **3**

Question 3 (12 marks) Begin a new booklet**Marks**

(a) If $5x^3 - 6x^2 - 29x + 6 = 0$ has roots α, β, γ then find the value of $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$ **2**

(b) Use Mathematical Induction to prove that $2^n \geq 1 + n$ for $n \geq 1$ **4**

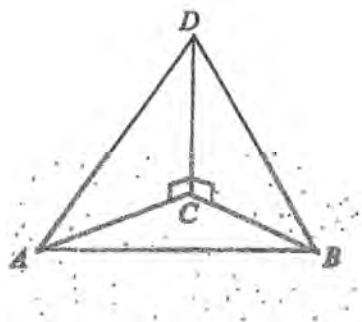
(c) Use the substitution $u = x^2 + 1$ to evaluate $\int_1^7 \frac{x}{(1+x^2)^2} dx$. **4**

(d) Find the sum of the infinite series $\sin^2 x + \sin^4 x + \sin^6 x + \dots$. Express your answer in simplest form. **2**

Question 4 (12 marks) Begin a new booklet

Marks

(a)



Three points A , B and C lie on a plane. Points A and B are 30 metres apart and $\angle ACB = 120^\circ$. A vertical flagpole, CD , of height h metres stands at C . From A the angle of elevation of the top, D , of the flagpole is 30° . From B the angle of elevation to D is 45° .

- | | |
|--|----------|
| (i) Find the length of AC and BC in terms of h . | 2 |
| (ii) Hence find the value of h to one decimal place. | 2 |

(b) The parametric equations of a parabola are

$$x = 2t$$

$$y = 2t^2$$

- | | |
|---|----------|
| (i) Find the Cartesian equation of the parabola. | 1 |
| (ii) Find the equation of the tangent to this parabola at the point where $x = 2$. | 2 |

(c) Find $\int \sin^2 2x \, dx$	2
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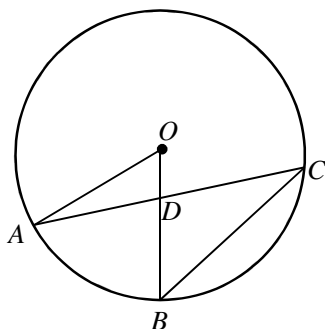
(d) (i) Find the domain and range of the function $f(x) = \cos^{-1}(2x)$.	2
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(ii) Sketch the graph of the curve $f(x) = \cos^{-1}(2x)$.	1
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Question 5 (12 marks) Begin a new booklet

Marks

(a)



A , B and C are points lying on circle with centre O . AO is parallel to BC .
 OB and AC intersect at D . $\angle ACB = 31^\circ$.

(i) Find the size of $\angle AOB$, giving reasons. **2**

(ii) Find the size of $\angle BDC$, giving reasons. **2**

(b) At time t years after the start of the year 2000, the number of individuals in A population is given by $N = 80 + Ae^{0.1t}$ for some constant $A > 0$.

(i) Show that $\frac{dN}{dt} = 0.1(N - 80)$. **1**

(ii) If there were 100 individuals in the population at the start of the year 2000
 Find the year in which the population size is expected to reach 200. **3**

(c) (i) Write $\sqrt{3} \cos x - \sin x$ in the form $r \cos(x + \alpha)$. **1**

(ii) Hence, give the general solution to $\sqrt{3} \cos x - \sin x = 1$ **3**

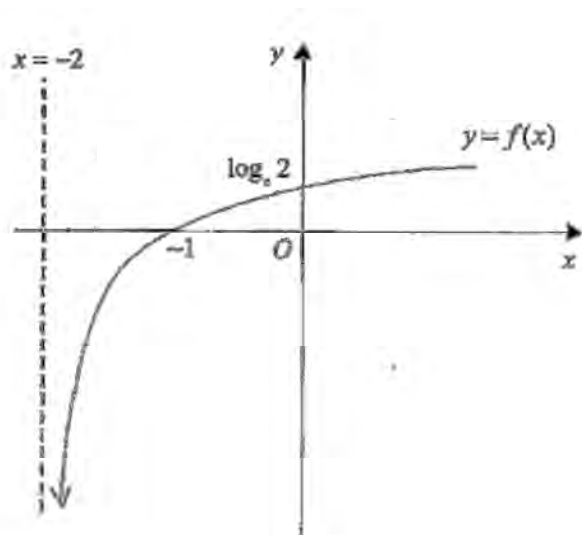
Question 6 (12 marks) Begin a new booklet

Marks

- (a) A particle is moving in a straight line. At time t seconds it has displacement x metres from a fixed point O in the line, $v \text{ ms}^{-1}$ is given by $v = \frac{1}{x+1}$ and acceleration $a \text{ ms}^{-2}$. Initially the particle is at O .

- (i) Express a as a function of x . **1**
- (ii) Express x as a function of t . **3**

(b)



The diagram shows the graph of the function $f(x) = \ln(x+2)$.

- (i) Copy the diagram and on it draw the graph of the inverse function $f^{-1}(x)$ showing the intercepts on the axes and the equation of the asymptote. **2**
- (ii) Show that the x -coordinates of the points of intersection of the curves $y = f(x)$ and $y = f^{-1}(x)$ satisfy the equation $e^x - x - 2 = 0$. **2**
- (iii) Show that the equation $e^x - x - 2 = 0$ has a root α such that $1 < \alpha < 2$. **2**
- (iv) Use one application of Newton's method with an initial approximation $\alpha_0 = 1.2$ to find the next approximation for the value of α , giving your answer correct to one decimal place. **2**

Question 7 (12 marks) Begin a new booklet

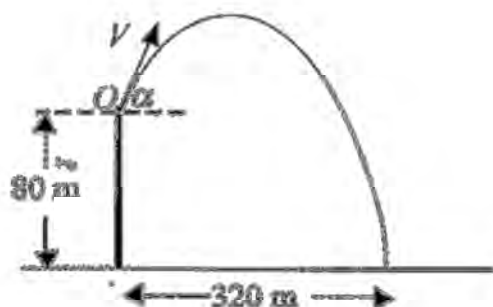
Marks

- (a) A particle is moving in a straight line with Simple Harmonic Motion. At time t seconds it has displacement x metres from a fixed point O on the line, given by $x = 1 + 3 \cos \frac{t}{2}$, velocity $v \text{ ms}^{-1}$ and acceleration $a \text{ ms}^{-2}$.

(i) Show that $a = -\frac{1}{4}(x-1)$. **2**

- (ii) Find the distance travelled and the time taken by the particle over one complete oscillation of its motion. **2**

(c)



A particle is projected with speed $V \text{ ms}^{-1}$ at an angle α above the horizontal from a point O at the top edge of a vertical cliff which is 80 m above horizontal ground. The particle moves in a vertical plane under gravity where the acceleration due to gravity is 10 ms^{-2} . It reaches its greatest height after 3 seconds and hits the ground at a horizontal distance of 320 m from the foot of the cliff.

The horizontal and vertical displacements, x and y metres respectively, of the particle from the point O after t seconds are given by $x = Vt \cos \alpha$ and $y = -5t^2 + Vt \sin \alpha$. (Do NOT prove these results.)

- (i) Show that $V \sin \alpha = 30$. **1**
- (ii) Show that the particle hits the ground after 8 seconds. **2**
- (iii) Show that $V \cos \alpha = 40$. **1**
- (iv) Hence find the exact value of V and the value of α to the nearest minute. **2**
- (v) Find the time after projection when the direction of motion of the particle first makes an angle of 45° below the horizontal. **2**

EXT SOLS

Q1 (a) $2(2x-1)(4x^2+2x+1)$

(2)

(b) $\frac{2(6)-3}{3}, \frac{-4(2)+2(1)}{3}$

$P(3, -2)$

(2)

(c) $\frac{2}{x+1} < 1$

$2x+2 < x^2+2x+1$

$0 < x^2-1$



$x < -1$ or $x > 1$

(2)

(d) $f'(x) = 27x^2$ and $x^2 > 0$
so $27x^2$ is always > 0
 $\therefore$ pos. gradient
always increasing

(2)

(e) (i) $u=x$ $v=\cos^2 3x$
 $u'=1$ $v'=-6\cos 3x \sin 3x$

$\therefore \frac{d}{dx}(x\cos^2 3x) = \cos^2 3x - 6x\cos 3x \sin 3x$

(2)

(ii) $u=x^2$ $v=\tan^{-1} x$
 $u'=2x$ $v'=\frac{1}{1+x^2}$

$2x\tan^{-1} x + \frac{x^2}{1+x^2}$

(2)

Q2

(a) $P(2) = 2^3 - 2(2)^2 + k(2) + k^2$

$$0 = 8 - 8 + 2k + k^2$$

$$0 = k(2+k)$$

$$k = 0, -2.$$

(2)

(b) $y = 3x + 1 \quad m_1 = 3$

$$y = -x + 5 = 0 \quad m_2 = -1$$

$$\tan \theta = \left| \frac{3 - (-1)}{1 \times 3(-1)} \right| = 2$$

$$\therefore \theta \doteq 63^\circ$$

(2)

(c) $\int_0^1 \frac{dx}{\sqrt{4-x^2}} = \left[\sin^{-1} \frac{x}{2} \right]_0^1$

$$= \sin^{-1} \left(\frac{1}{2} \right) - \sin^{-1}(0)$$

$$= \frac{\pi}{6} - 0$$

$$= \frac{\pi}{6}$$

(2)

(d) $\left(1 - \frac{1-t^2}{1+t^2} \right) \div \frac{2t}{1+t^2} \quad \textcircled{1}$

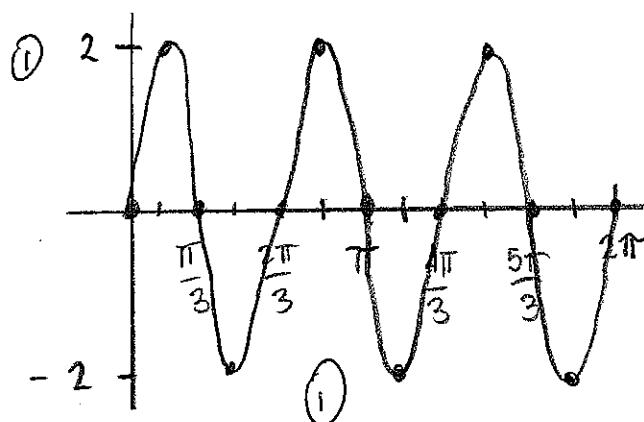
$$\frac{1+t^2 - 1+t^2}{1+t^2} \times \frac{\cancel{1+t^2}}{2t} \quad \textcircled{1}$$

$$\frac{2t^2}{2t} = t$$

$$\text{but } t = \tan \frac{\alpha}{2} \quad \textcircled{1}$$

(e) $y = 2 \sin 3x \quad \frac{2\pi}{3} = \text{per} \quad \textcircled{1}$

Total
(3)



Q3

(a) $\alpha + \beta + \gamma = \frac{-b}{a}$

$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$

$\alpha\beta\gamma = \frac{-d}{a}$

$\frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma} = \frac{\frac{c}{a}}{\frac{-d}{a}}$

$\frac{c}{a} \times \frac{a}{-d}$

$-\frac{29}{5} \times \frac{5}{-6} = \left(\frac{29}{6}\right) \quad (2)$

(b) Show true for $n=1$

$2^1 \geq 1+1$

$2 \geq 2 \quad \checkmark$

$\therefore$ true for $n=1$

Assume true for $n=k$

$2^k \geq 1+k$

Prove true for $n=k+1$

LHS $= 2^{k+1} \geq \dots$ RHS $= k+1+1 = k+2$

$2^k \cdot 2$

but $2^k \geq 1+k$
 $\therefore 2 \times 2^k \geq 2(1+k) = 2+2k$ which is greater than $k+2$ if $k \geq 1$

(4)

c) $u = x^2 + 1$

$\frac{du}{dx} = 2x$
 $du = 2x dx$

$\therefore \frac{1}{2} \int_2^{50} \frac{du}{u^2} = -\frac{1}{2} \left[\frac{1}{u} \right]_2^{50}$
 $= -\frac{1}{2} \left[\frac{1}{50} - \frac{1}{2} \right]$
 $= \frac{6}{25} = 0.24$

(4)

d)

$S_n = \frac{a}{1-r}$

$= \frac{\sin^2 x}{1 - \sin^2 x}$

$= \frac{\sin^2 x}{\cos^2 x}$

$= \tan^2 x \quad 2$

$$4a) \frac{AC}{h} = \tan 60^\circ$$

$$AC = h \tan 60$$

$$AC = h\sqrt{3}$$

$$BC = h \tan 45^\circ$$

$$BC = h$$

$$900 = 3h^2 + h^2 - 2\sqrt{3}h^2 \cos 120^\circ$$

$$900 = h^2 (3 + 1 - 2\sqrt{3} \times -1/2)$$

$$900 = h^2 (5.732 \dots)$$

$$157.01 \dots = h^2$$

$$h \doteq 12.53$$

$$\sqrt{3} \cos x - \sin x = r (\cos x \cos \alpha - \sin x \sin \alpha)$$

$$r = \sqrt{\sqrt{3}^2 + 1^2} = \sqrt{4} = 2$$

$$2 \cos \alpha = \sqrt{3}$$

$$2 \sin \alpha = 1$$

$$\cos \alpha = \frac{\sqrt{3}}{2}$$

$$\sin \alpha = 1/2$$

$$\tan \alpha = \frac{1}{\sqrt{3}}$$

$$\alpha = 30^\circ$$

$$2 \cos(x + 30^\circ) = 1$$

$$\cos(x + 30^\circ) = 1/2$$

$$\cos(x + 30^\circ) = \cos(60^\circ)$$

$$x + 30 = 360n \pm 60^\circ$$

$$x = 360n + 30^\circ \quad \text{or} \quad 360n - 90^\circ$$

4b)

$$x = 2t$$

$$y = 2t^2$$

(i)

$$\frac{x}{2} = t$$

$$\therefore y = 2 \left(\frac{x}{2} \right)^2$$

$$y = 2 \left(\frac{x^2}{4} \right)$$

$$y = \frac{x^2}{2}$$

$$2y = x^2$$

(1)

(ii)

$$y' = \frac{1}{2} \times 2x$$

$$= x$$

$$\textcircled{a} \quad x = 2 \quad m_t = 2$$

$$\textcircled{a} \quad x = 2 \quad y = 2$$

$\therefore$ equation of tangent is

$$y - 2 = 2(x - 2)$$

$$y - 2 = 2x - 4$$

$$y = 2x - 2$$

(2)

$$(c) \int \sin^2 2x \, dx$$

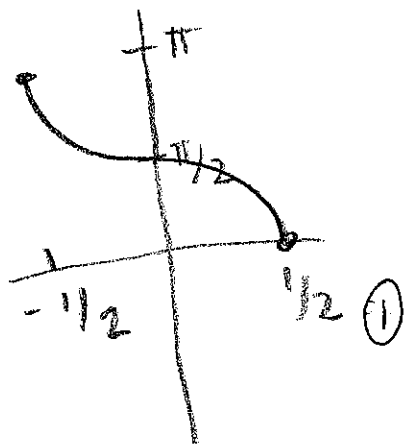
$$= \frac{1}{2} \int (1 - \cos 4x) \, dx$$

$$= \frac{1}{2} x - \frac{1}{2} \times \frac{1}{2} \sin 4x + C$$

$$= \frac{1}{2} x - \frac{1}{4} \sin 4x + C$$

(2)

Q4) d) (ii)



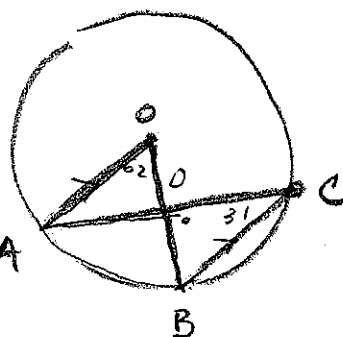
(i) $D: -\frac{1}{2} \leq x \leq \frac{1}{2}$
 $R: 0 \leq y \leq \pi$. (2)

Q5)

(i) $\angle AOB = 62^\circ$ angle at the centre is twice the angle on circumference standing on the same arc. (2)

(ii) $\angle CBD = 62^\circ$ (alt angles bet parallel lines)

and $\therefore \angle CDB = 87^\circ$ (angle sum of Δ)



(b) (i) $N = 80 + Ae^{0.1t}$

$$\frac{dN}{dt} = 0.1 Ae^{0.1t} \quad (1)$$

$$= 0.1(N - 80)$$

(ii) $N = 100$ when $t = 0 \therefore A = 20$

$$200 = 80 + 20e^{0.1t}$$

$$e^{0.1t} = 6$$

$$0.1t = \ln 6$$

$$t = 10 \ln 6$$

$$\therefore t = 17.92$$

So in 2017 the pop'n reaches 200.

Q6)

a i) $v = \frac{1}{2x+1}$

$$\frac{d}{dx} \left(\frac{1}{2(x+1)^2} \right) = a$$

$$= \frac{-1}{(x+1)^3}$$

(ii) $\frac{dx}{dt} = \frac{1}{x+1}$

$\therefore \frac{dt}{dx} = x+1$

$$t = \int (x+1) dx$$

$$= x^2 + x + C$$

when $t=0$ $x=0$

$\therefore C=0$

$$t = x^2 + x$$

~~$t = x^2$~~

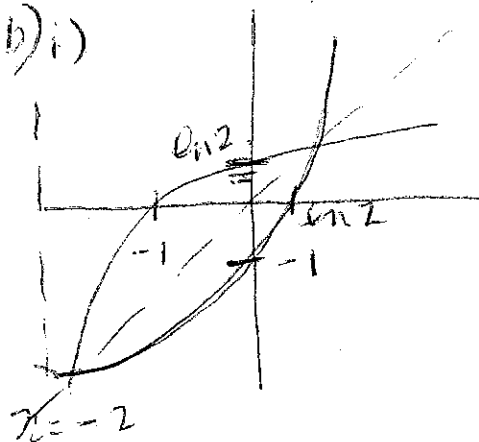
$$t + \frac{1}{4} = x^2 + x + \frac{1}{4}$$

$$t + \frac{1}{4} = \left(x + \frac{1}{2}\right)^2$$

$$\pm \sqrt{t + \frac{1}{4}} = x + \frac{1}{2}$$

$$\therefore -\frac{1}{2} \pm \sqrt{t + \frac{1}{4}} = x$$

(b) i)



(2)

(ii) intersect where $y=x$

$$\ln(x+2) = x$$

$$x+2 = e^x$$

$$e^x - x - 2 = 0 \quad \checkmark \quad (2)$$

(iii)

$$g(x) = e^x - x - 2$$

$$g(1) = e - 3 < 0 \quad \text{and}$$

$$g(2) = e^2 - 4 > 0$$

$$\therefore 1 < x < 2 \quad \text{where}$$

x is a root.

(iv)

$$g'(x) = e^x - 1$$

$$1.2 = \frac{e^{1.2} - 3.2}{e^{1.2} - 1} \approx (1.1)$$

(2)

7a) $x = 1 + 3 \cos \frac{t}{2}$, v m/sec a m/sec²
 (i)

$$v = \frac{dx}{dt} = -\frac{3}{2} \sin \frac{t}{2}$$

$$a = \frac{d}{dt} \left(\frac{dx}{dt} \right) = -\frac{3}{4} \cos \frac{t}{2}$$

but $x - 1 = 3 \cos \frac{t}{2}$

(2)

$$\therefore a = -\frac{1}{4} (x-1) v$$

(ii) $v = 0$ when $-\frac{3}{2} \sin \frac{t}{2} = 0$

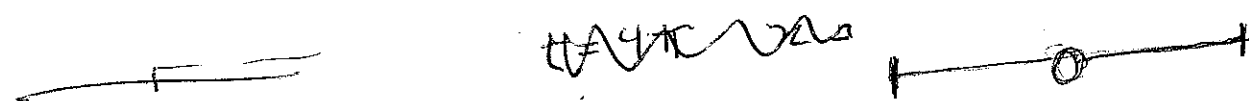
at rest

$$\sin \frac{t}{2} = 0$$

$$t = 0, t = 2\pi, t = 4\pi \dots$$

$$\therefore \text{when } t = 0 \quad x = 1 + 3 = 4$$

$$t = 2\pi \quad x = 1 - 3 = -2$$



$\therefore$ from 0 ~~for~~ and back is 12m.

and it takes 4π sec

$$\text{period } \frac{2\pi}{1/2} = \textcircled{4\pi}$$

b.

(i) $\dot{y} = -10t + V \sin \alpha$ and $\dot{y} = 0$ when $t = 3$

$$0 = -30 + V \sin \alpha$$

$$\therefore V \sin \alpha = 30$$

(ii) $y = -80 = -5t^2 + 30t$

$$\therefore t^2 - 6t - 16 = 0 \quad (t \geq 0)$$

$$(t-8)(t+2) = 0 \quad \therefore t = 8$$

(iii) $x = 320$ when $t = 8$

$$320 = 8V \cos \alpha$$

$$\therefore V \cos \alpha = 40$$

(iv) $V^2 (\sin^2 \alpha + \cos^2 \alpha) = 30^2 + 40^2$

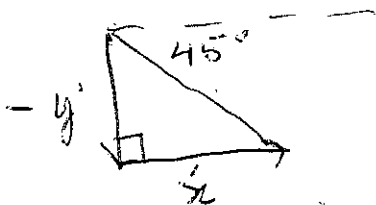
$$V^2 = 2500$$

$$\therefore V = 50$$

$$\frac{V \sin \alpha}{V \cos \alpha} = \frac{30}{40} \quad \therefore \tan \alpha = \frac{3}{4}$$

$$\text{and } \alpha = 36^\circ 52'$$

(v)



$$\dot{y} = -\dot{x}$$

$$-10t + 30 = -40$$

$$\therefore 10t = 70$$

$$t = 7$$

hence after 7 seconds